

Rachana Venati

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Experience

Working Student | Dec 2024 – Presents

Kontacts IT-Solutions GmbH, Potsdam

- Developed and deployed a **RAG** system on **Netcup** and **GCP**, integrating **vector search**, **keyword search**, **reranking**, and **agentic** workflows to improve document based troubleshooting.
- Built chatbot PoCs to evaluate **late chunking**, **semantic chunking**, **embedding** models, and **retrieval techniques**, securing approval for further development with the Hasso Plattner Institute.
- **Automated** customer order notifications and document updates using **n8n** and **Activepieces**, reducing manual operational effort.

Software Engineer II | Jul 2020 – Sep 2023

HCL Software, India

- **Resolved** more than **250 critical defects** across the **iOS**, **Android**, and **web** versions of the **VoltMX** product.
- Supported customer **push notification** systems using **GCM**, **FCM**, and **APNs** during peak business-load periods.
- Conducted **team enablement** and **knowledge transfer** sessions for newly onboarded teams across the United States and Japan.

Software Engineer | Sep 2019 – Jul 2020

Kony, Inc | Temenos, India

- Troubleshoot and resolved Android product issues, with 80% of cases completed within strict SLA requirements.
- Contributed actively to the customer portal by publishing technical articles and PoCs covering release features and complex native Android application integrations.

Master's Thesis & Projects

Experimental Evaluation of Axioms for Health Information Retrieval

Master's thesis

- Designed and implemented seven explainable axioms for health information retrieval, covering medical terminology, age, sex, recency, and evidence level.
- Built pairwise document-preference functions using MeSH-normalized medical concepts, PubMed metadata, lexicon-based demographic matching.
- Conducted post hoc and example-based analyses to compare axiom preferences with expert utility judgments and identify failures in neural retrieval rankings.
- Developed an axiomatic re-ranking pipeline using majority voting and KwikSort to convert pairwise preferences into complete document rankings.
- Evaluated the approach on the best runs submitted to TREC 2021 Health Misinformation, CLEF 2017 eHealth.
- Nearly 90% of axiom preferences, aligned with expert utility judgments more consistently than with the best retrieval runs from the evaluated tracks.
- Reranking the best run submitted to TREC 2021 using axioms improved nDCG@5 from 0.492 to 0.524.
- Reranking the best run submitted to CLEF 2017 improved all reported metrics: nDCG@5 from 0.119 to 0.125, P@5 from 0.159 to 0.165, and RBP(0.8) from 0.148 to 0.150.

Nutri score (MultiModal Instagram post classification)
Special Project

- Collected 825 multimodal Instagram posts containing captions and images using Zeeschuimer and 4CAT.
- Annotated the dataset in Label Studio with positive, negative, and neutral sentiment labels across text, image, and combined modalities, including hashtag and emoji analysis.
- Fine-tuned Twitter-RoBERTa on Instagram caption data, improving precision and recall by at least 22%.
- Trained a Vision Transformer on Instagram images and developed an early-fusion model by combining text and image embeddings through a feed-forward network.
- Achieved at least 16% higher precision and recall than the unimodal models and built a FastAPI application for local inference.

Retrival Agumented Generation
Special Project

- Built a TREC 2024 retrieval pipeline for the MS MARCO dataset, generating embeddings and indexing over 138 million segmented passages in Weaviate vector database.
- Implemented and compared Keyword, Semantic, and Hybrid retrieval strategies.
- Evaluated retrieval performance using nDCG, retrieval results outperformed baseline provided by the track.
- Developed a Llama 3-based generation pipeline that produced answers for retrieved queries and cited supporting input passages at sentence level using post-processing and clustering.

Education

Master of Science in Computer Science for Digital Media <i>Bauhaus-Universität Weimar, Germany</i>	Oct 2023 – Jun 2026 GPA: 1.5
Bachelor of Technology in Information Technology <i>Anna University, India</i>	Aug 2016 – Apr 2019 GPA: 1.7

Skills

Programming: Python, Javascript,C#
AI & NLP: RAG, Prompt Engineering, Fine-tuning
Frameworks & Tools: Transformers, LangGraph, PyTorch, Pydantic, Scikit-learn, Docker, FastAPI, SLURM.

Languages

English	C1
German	B1
